조명래 교수

연구분야

저차원 물질, 극저온 고자기장 실험, DC-RF 전자 수송특성, 나노소자 공정, 초전도 큐빗, 양자홀

KAIST 양자대학원 참여교수

반도체특성화대학 사업단 참여교수: 담당과목. 양자나노반도체, PBL 기반 반도체 특성화 프로젝트

개인 과제

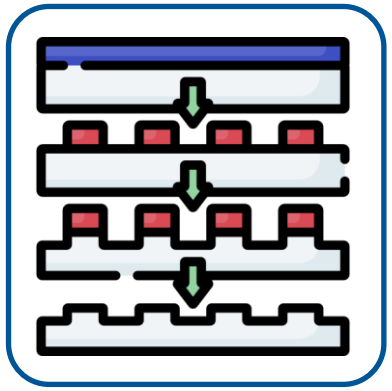
우수 신진연구 ”2차원 양자물질을 이용한 전자 양자광학”

국가간협력기반조성사업 “양자전자소자 연구를 위한 한-EU 국제 협력”

신진연구자 인프라 지원사업 “양자 전자 소자 연구 수행을 위한 저온 냉동기 구축”

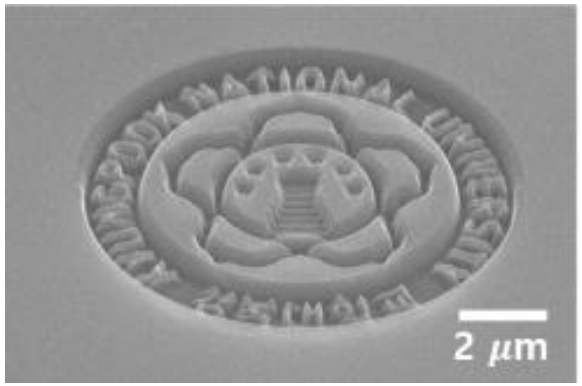
작은 구조를 만들고, 그 동작을 측정한다, 극단적인 환경에서, 양자적 현상이 보이게?

Nanofabrication

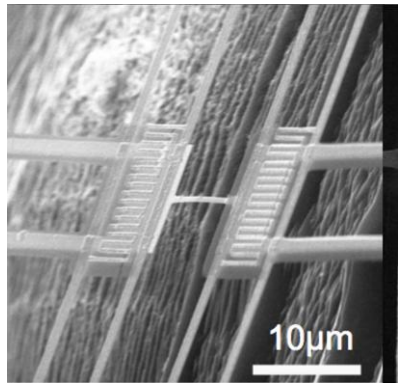


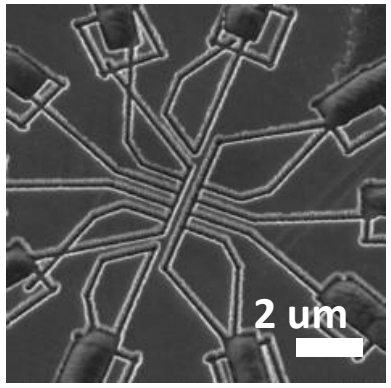
- Smaller the better, electron cannot travel far with its exotic quantumness!
- Techniques and knowledge relevant to semiconductor industry
- Lithography, Etching, Deposition, Packaging…

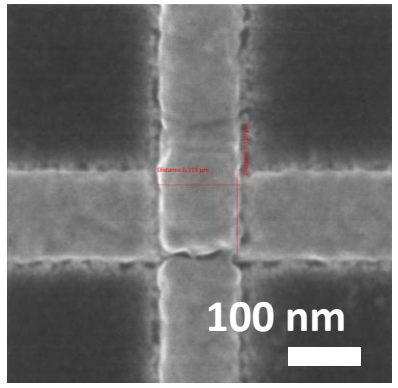
조셉슨정션
NEMS
Quantum dot…

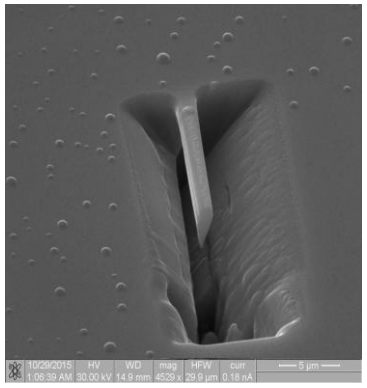


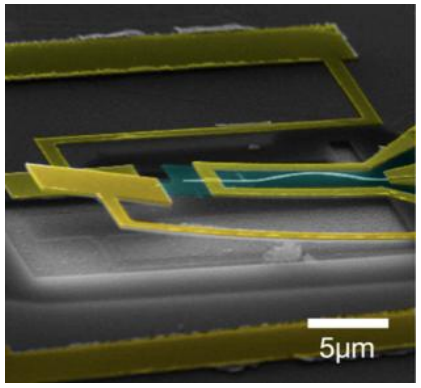




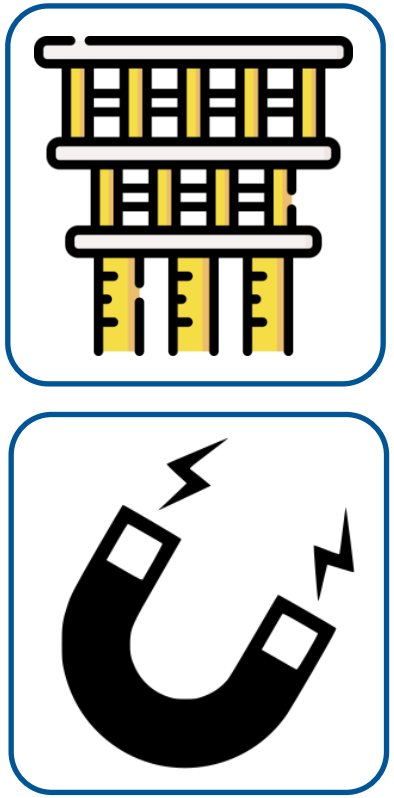


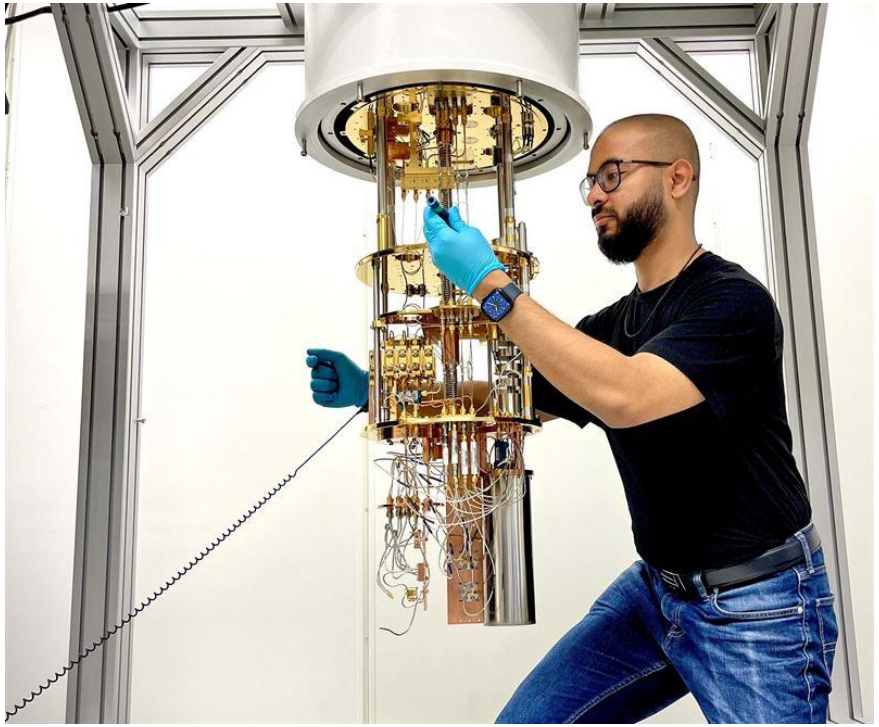




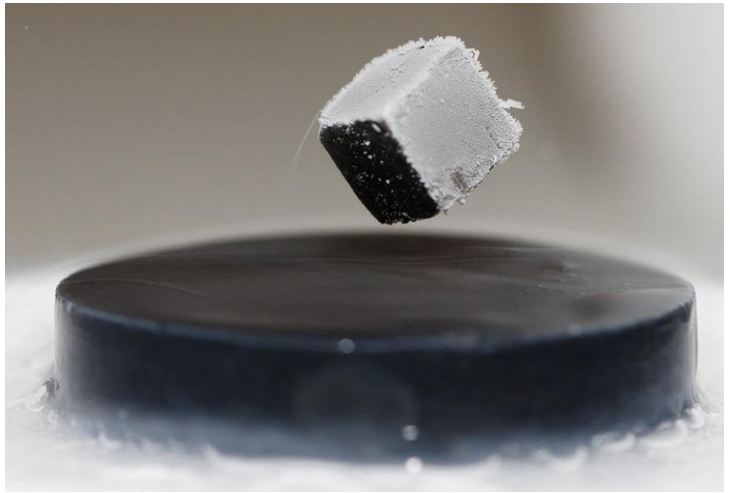


Low temperature and high magnetic field





희석식 냉동기



초전도체

- A technique can cool down handful of material to 10mK (1/100 K above absolute Zero)
- Superconducting magnet can generate more up to 14T continuously (MRI 0.5 – 3T, 1x1 inches Neodymium magnet surface <1 T)

사진 Chalmers univ. of Tech.